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RESEARCH ARTICLE

Structure of the Parasite Infracommunity of *Hoplobatrachus tigerinus* Daudin, 1803 from YSR Kadapa District, Andhra Pradesh, India

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Abstract Three hundred specimens of *Hoplobatrachus tigerinus* (Daudin, 1803), 133 females and 167 males, collected in YSR Kadapa district, Andhra Pradesh, India, from 2013 to 2015, were examined for the presence of metazoan parasites. One hundred seventy-two (57.3%) frogs were parasitized by at least one or more metazoan parasite species. Nine species of parasites were collected: 6 digeneans, 1 cestode and 2 nematodes. All the recovered parasites are satellite species with less than 33% prevalence. Digeneans are the most abundant group of parasites accounting for 89.9% of total parasites. *Hoplobatrachus tigerinus* is a new host record for *Paracephalogonimus minutum*. *Paracephalogonimus minutum* is the dominant species followed by *Tremiorchis ranarum*. The metazoan parasite species of *H. tigerinus* showed the typical aggregated pattern of distribution of most parasite systems. Only *Cosmocercoides variabilis–Oswaldocruzia filiformis* (0.23), *Paracephalogonimus minutum—Ophiotaenia tigrina* (0.13), *Diplodiscus amphichrus–Ohiotaenia tigrina* (0.13), *Cosmocercoides variabilis–Mehraorchis ranarum* (0.11) and *Ganeo tigrnum–Diplodiscus amphichrus* (0.11) showed significant interspecific associations while the other pairs showed very less to no association. There was a very negligible influence of host snout-vent length on the parasitism however; there is no influence of host sex on the parasitic abundance.

Keywords *Hoplobatrachus tigerinus* · Metazoan parasites · Community structure

Introduction

Parasitic communities are hierarchial and usually a host presents more or less uniform type of parasite assemblage throughout the world (Ulmer 1970; Holmes and Price 1986; Esch and Fernandez 1993; Sousa 1994; Poulin 2001; Dambacher 2002). Interactive versus isolationist classification of parasite communities has been the most successful theoretical framework for parasite community ecology research (Holmes and Price 1986; Esch et al. 1990; Sousa 1994; Poulin 2001; Poulin and Luque 2003). The parasite communities of amphibians are deemed to be extremely unpredictable, depauperate and non-interactive (Aho 1990). There are several areas in India, in particular in Southern India especially YSR Kadapa where the biodiversity of metazoan parasites in vertebrate hosts are poorly investigated. Amphibians are abundantly found in the study area due to the warm and humid climates of YSR Kadapa District, Andhra Pradesh. These hosts serve as the most preferred vertebrate hosts for a number of metazoan endoparasites like digeneans, cestodes and nematodes. *Hoplobatrachus tigerinus* Daudin, 1803 (Dicroglossidae Anderson, 1871) is one of the prevalent and recurrently occurring aquatic species of amphibians of YSR Kadapa District which dwells in the low lying areas like ponds, ditches and irrigated fields. The number of metazoan parasites that a host species supports varies widely from host to host and from location to location (Price and Clancy 1983). Also, the amphibians are one of the vertebrate hosts affected by the anthropogenic activities which might decrease or increase the populations of parasites in the

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